

```
In [87]: import cv2
import re
from matplotlib import pyplot as plt
import numpy as np
import pytesseract
import pymongo
from PIL import Image
im_file = "C:/Users/neeta/temp/nvc.jpg"
img = cv2.imread(im_file)
```

```
In [88]: def display(im_path):
    dpi = 80
    im_data = plt.imread(im_path)
    height, width = im_data.shape[:2]
    figsize = width / float(dpi), height / float(dpi)
    fig = plt.figure(figsize=figsize)
    ax = fig.add_axes([0, 0, 1, 1])
    ax.axis('off')
    ax.imshow(im_data, cmap='gray')
    plt.show()
display(im_file)
```



भारत निर्वाचन आयोग
ELECTION COMMISSION OF INDIA
मतदाता फोटो पहचान पत्र - Elector Photo Identity Card

UR [REDACTED] 60



नाम: [REDACTED] द्विवेदी

Name: [REDACTED] Dwivedi

पिता का नाम: [REDACTED] कुमार द्विवेदी

Father's Name: [REDACTED] Kumar Dwivedi

लिंग / Gender: पुरुष / Male

जन्म तिथि / आयु:

Date of Birth / Age: [REDACTED]-2004



```
In [89]: def grayscale(image):  
         return cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
In [90]: gray_image = grayscale(img)  
         cv2.imwrite("C:/Users/neeta/temp/graya2.jpg", gray_image)
```

```
Out[90]: True
```

```
In [91]: display("C:/Users/neeta/temp/graya2.jpg")
```



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```
In [1]: def getSkewAngle(cvImage) -> float:
        newImage = cvImage.copy()
        blur = cv2.GaussianBlur(newImage, (9, 9), 0)
        thresh = cv2.threshold(blur, 0, 255, cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU)[1]

        kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (30, 5))
        dilate = cv2.dilate(thresh, kernel, iterations=5)

        contours, hierarchy = cv2.findContours(dilate, cv2.RETR_LIST, cv2.CHAIN_APPROX_
        contours = sorted(contours, key = cv2.contourArea, reverse = True)

        largestContour = contours[0]
        minAreaRect = cv2.minAreaRect(largestContour)

        angle = minAreaRect[-1]
        if angle < -45:
            angle = 90 + angle
        return -5.0 * angle
```

```
In [2]: def rotateImage(cvImage, angle: float):
        newImage = cvImage.copy()
        (h, w) = newImage.shape[:2]
        center = (w // 2, h // 2)
        M = cv2.getRotationMatrix2D(center, angle, 1.0)
        newImage = cv2.warpAffine(newImage, M, (w, h), flags=cv2.INTER_CUBIC, borderMod
        return newImage

        def deskew(cvImage):
            angle = getSkewAngle(cvImage)
            return rotateImage(cvImage, -1.0 * angle)
```

```
In [94]: fixim = deskew(gray_image)
        cv2.imwrite("C:/Users/neeta/temp/rotfix.jpg", fixim)
        display("C:/Users/neeta/temp/rotfix.jpg")
```



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Date of Birth / Age: [REDACTED] 2004



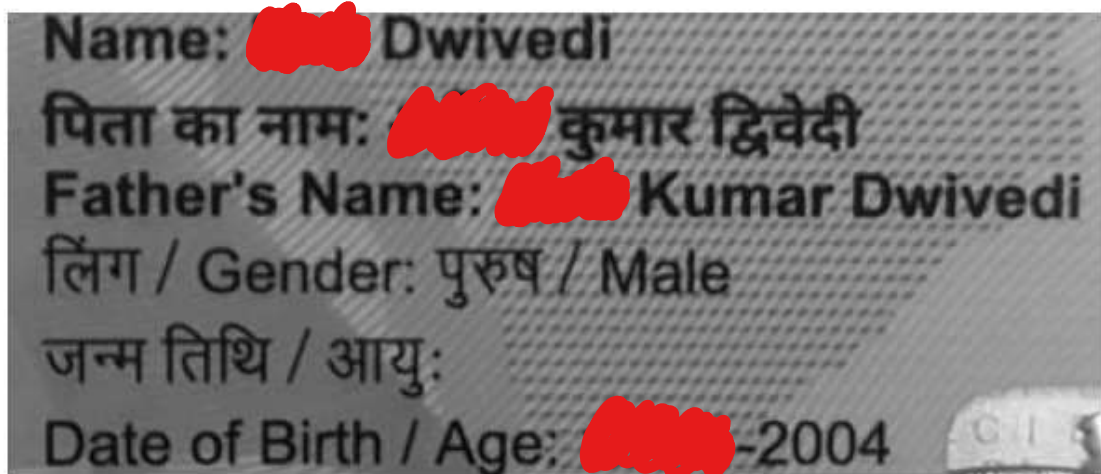
```
In [95]: print(img.shape)
```

```
(1280, 720, 3)
```

```
In [96]: cropped_image1 = fixim[620:820, 220:690]  
cv2.imwrite("C:/Users/neeta/temp/CropAd.jpg", cropped_image1)
```

```
Out[96]: True
```

```
In [97]: display("C:/Users/neeta/temp/CropAd.jpg")
```



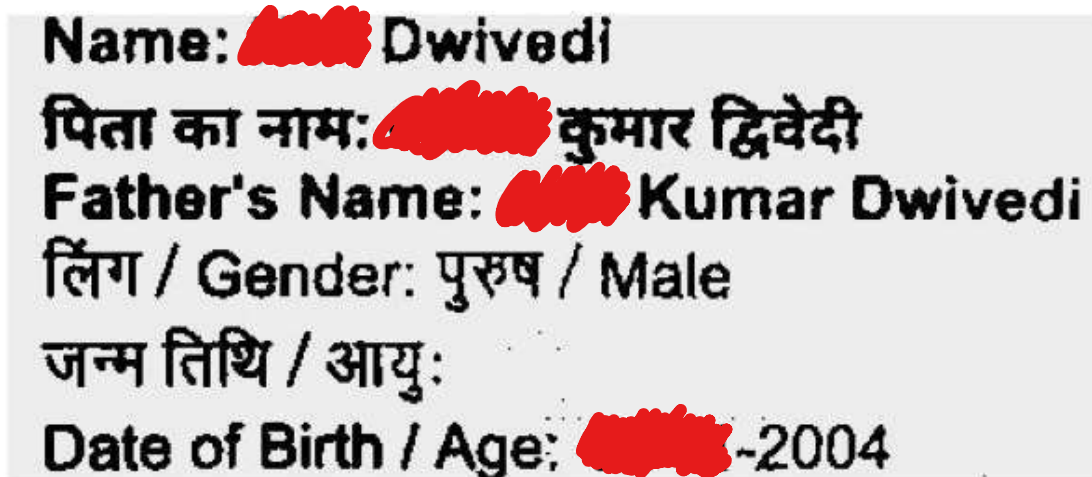
```
In [98]: cropped_image2 = fixim[530:580, :175]  
cv2.imwrite("C:/Users/neeta/temp/CropAdd.jpg", cropped_image2)
```

```
Out[98]: True
```

```
In [99]: display("C:/Users/neeta/temp/CropAdd.jpg")
```



```
In [105... thresh, im_bw1 = cv2.threshold(cropped_image1, 90, 100, cv2.THRESH_BINARY)  
cv2.imwrite("C:/Users/neeta/temp/bw1.jpg", im_bw1)  
display("C:/Users/neeta/temp/bw1.jpg")
```



```
In [107... thresh, im_bww2 = cv2.threshold(cropped_image2, 98, 100, cv2.THRESH_BINARY)
cv2.imwrite("C:/Users/neeta/temp/bww2.jpg", im_bww2)
display("C:/Users/neeta/temp/bww2.jpg")
```



```
In [108... im_file = "C:/Users/neeta/temp/bww1.jpg"
```

```
In [109... img = Image.open(im_file)
ocr_resultf = pytesseract.image_to_string(img)
```

```
In [110... print(ocr_resultf)
```

```
Name: [REDACTED] Dwivedi

fen ar art: orfrat epee feat
Father's Name: [REDACTED] Kumar Dwivedi
fer / Gender: $89 / Male

aH fete / any:

Date of Birth / Age: [REDACTED]-2004
```

```
In [111... im_fileb = "C:/Users/neeta/temp/bww2.jpg"
```

```
In [113... image = Image.open(im_fileb)
ocr_resultg = pytesseract.image_to_string(image)
```

```
In [114... print(ocr_resultg)
```

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UR [REDACTED] 60
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In [115... pattern = re.compile('\d.*\d')
matches = pattern.finditer(ocr_resultf)
```

```
In [116... for match in matches:
    print(match)

<re.Match object; span=(98, 100), match='89'>
<re.Match object; span=(146, 156), match='[REDACTED]-2004'>
```

```
In [154... pattern = re.compile('(Male|..Male)\s')
matches = pattern.finditer(ocr_resultf)
```

```
In [155... for match in matches:
    print(match)

<re.Match object; span=(101, 108), match='/ Male\n'>
```

```
In [166... pattern = re.compile('\w.\s')
matches = pattern.finditer(ocr_resultf)
```

```
In [167... for match in matches:
    print(match)
```

```

<re.Match object; span=(0, 19), match='Name: [REDACTED] Dwivedi\n'>
<re.Match object; span=(20, 49), match='fen ar art: orfrat epee feat\n'>
<re.Match object; span=(49, 83), match='Father's Name: [REDACTED] Kumar Dwivedi\n'>
<re.Match object; span=(83, 108), match='fer / Gender: $89 / Male\n'>
<re.Match object; span=(109, 124), match='aH fete / any:\n'>
<re.Match object; span=(125, 157), match='Date of Birth / Age: [REDACTED]-2004\n'>

```

```

In [191... f1 = ocr_resultf[6:18]
f2 = ocr_resultf[64:82]
f3 = ocr_resultf[103:107]
f4 = ocr_resultf[146:156]
f5 = ocr_resultg[:10]

```

```

In [192... print(f1)
print(f2)
print(f3)
print(f4)
print(f5)

```

```

[REDACTED] Dwivedi
[REDACTED] Kumar Dwivedi
Male
[REDACTED]-2004
UR [REDACTED] 60

```

```

In [193... if __name__ == "__main__":
    client = pymongo.MongoClient("mongodb://localhost:27017")
    print(client)
    print(client.list_database_names())

```

```

MongoClient(host=['localhost:27017'], document_class=dict, tz_aware=False, connect
=True)
['OCR', 'admin', 'config', 'local', 'sample']

```

```

In [194... db = client['OCR']
collection = db['VoterID']

```

```

In [196... dic1 = {'Name': f1, "Father's Name" : f2, 'Sex' : f3, 'D.O.B.' : f4, 'VoterID N
collection.insert_one(dic1)

```

```

Out[196]: InsertOneResult(ObjectId('663f420f8edc6e1db2e98c51'), acknowledged=True)

```

```

In [197... print(dic1)

```

```

{'Name': '[REDACTED] Dwivedi', "Father's Name": '[REDACTED] Kumar Dwivedi', 'Sex': 'Male', 'D.
O.B.': '[REDACTED]-2004', 'VoterID Number': 'UR [REDACTED] 60', '_id': ObjectId('663f420f8edc
6e1db2e98c51')}

```